

Two engines. One vocal tuner.

Sonavyr is a pitch-correction plug-in that pulls vocals to the notes of your chosen key and scale in real time. Select the **Natural** engine for natural, transparent correction or **Synthetic** for sharp, modern hard-tune at the touch of a button.



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Audio
engines

17

DAW
parameters

10.7 ms

Synthetic
latency

±12

Semitone
transpose

AU · VST3
AAX

macOS 11+ ·
Apple Silicon
and Intel

What is Sonavyr?

Sonavyr is a pitch-correction plug-in for monophonic vocals: it continuously tracks pitch, finds the target note in the selected key and scale, and pulls the voice toward it at your chosen speed and amount. Two independent audio engines share one interface, key/scale selection, and Transpose / Detune controls.

NATURAL

Natural, transparent correction

- Adjustable amount and speed: gentle enough to preserve vibrato and slides, or sharp enough for a full "snap."
- Formant protection: preserves the corrected voice's character (timbre).
- ±12-semitone transposition; formants shift naturally with pitch.
- Independent Formant knob: shifts formants by -10 ... +10 semitones without changing pitch.
- The stereo image is preserved during transposition too.

SYNTHETIC

Live hard-tune

- Modern hard-tune / trap vocal character: sharp, clean transitions from note to note.
- 10.7 ms latency: suitable for monitoring during live performance and recording.
- Lock: how firmly the voice locks to the note. Snap: how quickly it reaches the note.
- Voices from 40 Hz to about 1650 Hz: from the lowest male rap to high female vocals.
- Both channels are processed at the same points; the stereo image is preserved.

Shared features

Key and scale

12 root notes; Chromatic, Major, Minor (natural), and Custom scales. In Custom, click notes on the ring to select them.

Transpose and Detune

Transpose -12 ... +12 semitones; Detune -50 ... +50 cents shifts the note grid's tuning (for tunings other than A4 = 440).

Latency compensation

Latency is reported precisely for every mode and setting; DAW delay compensation (PDC) aligns the audio with the dry track.

Two modes at a glance

	NATURAL	SYNTHETIC
Sound character	Natural, transparent	Modern hard-tune, robotic yet clean
Latency (Transpose 0 and below)	≈ 91-93 ms (Voice Range 80 Hz: ≈ 79-80 ms)	10.7 ms
Latency (Transpose +1 ... +12)	≈ 142-144 ms	22.8 ms
Detected pitch range	40-1500 Hz (80 Hz option)	40-≈ 1650 Hz
Sample rates (verified)	44.1 / 48 / 96 kHz	44.1 / 48 / 96 kHz
Dedicated controls	Amount, Speed + 7 advanced controls	Lock, Snap
CPU load (Apple M1, 48 kHz, avg.)	≈ 4.7%	≈ 3.2%
Best suited for	Recording, editing, and mixing	Live performance, recording, vocal FX

Natural correction, preserved character

Natural measures pitch every 5 ms, selects the target note stably, and shifts the voice by resampling it in the time domain. After correction, the voice's spectral envelope (formants) is processed separately, so the voice sounds natural rather than "shifted."

Pitch analysis

5 ms steps; 40 Hz (or 80 Hz)–1500 Hz; MPM with short-segment protection and octave-error correction.

Note decision

Target note in the key/scale; stable, evidence-weighted selection. Tolerance and Glide take effect here.

Correction curve

Approaches the note with the Speed time constant; Transition, Onset, Release, Amount.

Resynthesis

Cubic resampling + WSOLA (20 ms grains, 6 ms search); linked stereo.

Formant stage

Spectral envelope correction (Formant Protection) and formant layer (Formant).

Controls that shape correction

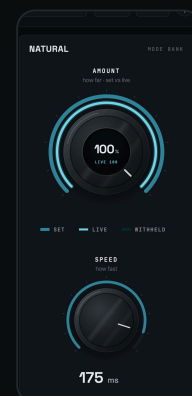
- **Amount** (0–100%): how much correction is applied. The knob's inner ring shows the correction actually being applied in real time.
- **Speed** (1–250 ms): the time constant for approaching the note. Full "snap" at 1 ms; gentle, musical correction at higher values.
- **Transition** (5–100% of Speed): the faster time constant used for 60 ms after a note change, settling quickly on the new note.
- **Glide** (0–100%): how much of a fast pitch slide is left uncorrected.
- **Onset** (5–200 ms) / **Release** (5–150 ms): how correction fades in at the start of each voiced segment and releases after the voice ends.
- **Tolerance** (0–50 cents): a zone around the note where no correction is applied, with a smooth transition to full correction outside it.

Formants and transposition

- **Formant Protection** (0–100%, default 100%): formant preservation during pitch correction. At 100%, the voice keeps its original character.
- **Formant** (–10 ... +10 semitones): shifts formants up or down without changing pitch for a darker/thicker or brighter/thinner tone. Uses cepstral envelope warping, a static bell filter, and level compensation at each value; active at Transpose 0 and adjustable during playback.
- **Transpose** (–12 ... +12): shifts the entire voice (including consonants); formants move with pitch, so the transposed voice sounds like a natural whole.
- **Voice Range**: "Full" goes down to 40 Hz; "80 Hz and above" reduces Natural's latency by about 12 ms.
- **Linked stereo**: stereo channels share processing points; the vocal stays in place in the stereo field.



Advanced view: Natural's fine-tuning panel



Amount (set / live / withheld) and Speed

Live hard-tune, low latency

Synthetic uses a low-latency, pitch-synchronous method (LE-PSOLA) that divides sound into pitch periods and rearranges them at the period markers. It produces sharp, clean note-to-note transitions without spectral smearing: the familiar character of modern hard-tune and trap vocals, with high audio quality.

Pitch detection

MPM, 1 ms control frame, 6.125 ms lookahead; separate detector below 50 Hz.

Target and lock

A note in the key/scale; Lock sets pull strength, Snap sets arrival speed.

Transpose / Detune

Added after target selection; note identity stays on the grid.

PSOLA synthesis

Period markers from both channels' average; both channels read at the same points.

Controls

- **Lock** (0–100%, default 100%): how strongly the voice is pulled to the note. 100% gives full hard-tune; lower values retain some natural pitch deviations.
- **Snap** (0–250 ms, default 0 ms): the time constant for reaching the note. 0 ms gives an instant jump (classic hard-tune); higher values give smoother transitions.
- **Key / Scale / Transpose / Detune**: shared with Natural.

Pitch range

Synthetic tracks voices from 50 Hz to approximately 1650 Hz; an extra detector handles very low voices at 40–50 Hz. This covers a wide range, from the lowest male rap recordings to high female vocals.

Latency

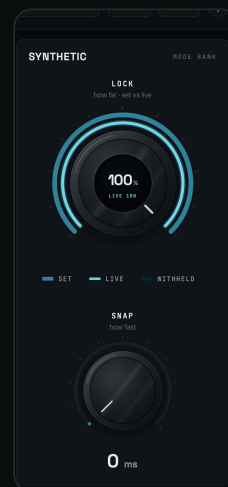
With Transpose at 0 or below, **10.7 ms** at every sample rate (472 samples at 44.1 kHz, 513 at 48 kHz, 1023 at 96 kHz); 22.8 ms at positive Transpose values. Key, scale, Lock, Snap and Detune do not affect latency.

Mode switching

Switching between Natural and Synthetic fades out the old engine in 5 ms. Once the new engine has buffered for its latency, it fades in over 5 ms, with no clicks. The DAW receives the new latency before the switch.



Synthetic mode, Standard view



Lock and Snap

17 DAW parameters

All parameters are saved with the session and, except Key / Scale, with presets. 14 parameters support DAW automation; the three that change latency (Mode, Transpose, Voice Range) cannot be automated, so the DAW's delay compensation remains intact during playback.

PARAMETER	MODE	RANGE	DEFAULT	AUTOMATION
Mode	—	Natural / Synthetic	Natural	No
Key	Shared	C ... B (12 roots)	C	Yes
Scale	Shared	Chromatic, Major, Minor (natural), Custom	Chromatic	Yes
Transpose	Shared	-12 ... +12 semitones	0	No
Detune	Shared	-50 ... +50 cents	0	Yes
Voice Range	Natural	Full (40 Hz) / 80 Hz and above	Full	No
Amount	Natural	0-100%	100%	Yes
Speed	Natural	1-250 ms	175 ms	Yes
Transition	Natural	5-100% of Speed	20%	Yes
Glide	Natural	0-100%	0%	Yes
Onset	Natural	5-200 ms	15 ms	Yes
Release	Natural	5-150 ms	30 ms	Yes
Tolerance	Natural	0-50 cents	0	Yes
Formant Protection	Natural	0-100%	100%	Yes
Formant	Natural	-10 ... +10 semitones	0	Yes
Lock	Synthetic	0-100%	100%	Yes
Snap	Synthetic	0-250 ms	0 ms	Yes

Non-parameter settings

- **Custom notes:** in the Custom scale, select notes by clicking or dragging on the note ring; an empty note set is not allowed.
- **Presets:** user presets are saved in ~/Library/Audio/Presets/StudioZIO/Sonavyr. Presets carry only the sound; key, scale and custom notes stay with the song.
- **A/B:** instantly compare two settings.
- **View and scale:** Simple / Standard / Advanced; 75%, 90%, 100%, 125%, 150%. Scales below the screen's minimum readability threshold are not shown.

Older sessions and presets can be opened; any new parameter absent from a saved session uses its default value.

Latency values

Sonavyr reports its exact latency to the DAW for every mode and setting. The DAW's delay compensation aligns processed audio with the dry track to sample accuracy. Values are in samples, with milliseconds in parentheses.

Natural

SETTING	44.1 kHz	48 kHz	96 kHz
Transpose 0 or below	4089 (92.7 ms)	4375 (91.1 ms)	8753 (91.2 ms)
Transpose +1 ... +12	6347 (143.9 ms)	6832 (142.3 ms)	13668 (142.4 ms)
Voice Range 80 Hz, Transpose 0 or below	3538 (80.2 ms)	3775 (78.6 ms)	7553 (78.7 ms)
Voice Range 80 Hz, Transpose +	5796 (131.4 ms)	6232 (129.8 ms)	12468 (129.9 ms)

Synthetic

SETTING	44.1 kHz	48 kHz	96 kHz
Transpose 0 or below	472 (10.7 ms)	513 (10.7 ms)	1023 (10.7 ms)
Transpose +1 ... +12	1006 (22.8 ms)	1095 (22.8 ms)	2187 (22.8 ms)

Voice Range does not affect Synthetic's latency.

How does latency change?

- **Key, scale, Amount, Speed, Lock, Snap, Detune, Formant** and all other sound controls do not affect latency; they can be adjusted freely during playback.
- **Downward Transpose** does not change latency and is heard immediately.
- **Upward Transpose** needs more lookahead. Above 0, the plugin switches once to the latency for +12; all values from +1 ... +12 are then heard immediately, even while dragging. Latency decreases again on returning to 0.
- These transitions briefly silence the audio while the engine buffers for its new latency (at 44.1 kHz, about 0.14 s in Natural when moving up, 0.09 s when returning to 0; about 0.02 s in Synthetic). The same applies to Mode and Voice Range changes. Transitions are click-free.
- The DAW always receives the new latency before the sound changes.

Validation

Latency values are extracted automatically from the product's own code and enforced by tests. Pro Tools' delay compensation display matched the table exactly in all 16 of 16 readings; REAPER's automated check in all 39 of 39. In Logic Pro, processed audio aligned to the new latency at every setting. In 44.1 / 48 / 96 kHz projects, processed audio aligned with the dry track (except for the known content shift with Natural transposition; see page 9).

10.7 ms

Synthetic · at every sample rate

≈ 79 ms

Natural · lowest with Voice Range 80 Hz

Correct pitch visually

The interface shows exactly what the engine is doing: the detected note, the target it is moving toward, and the pitch shift actually applied. Every displayed value comes from the engine's own analysis.

Note ring

- A 12-note ring: enabled scale notes, the detected note, and the target. In the Custom scale, click notes on the ring to edit them.
- The center shows the detected note, deviation in cents (SHARP / FLAT / IN TUNE), target, and applied shift.
- The needle, deviation arc, and correction arc update live; the precision indicator (loupe) on the outer ring magnifies small deviations.
- When transposed, the ring shows the Transpose 0 view rotated by the transposition amount (30° per semitone).



Transpose 0



Transpose -5: same view, rotated

Correction strip

Scrolling in analysis time: input pitch, target, shift, and intended output pitch on the note grid.

Amount ring

The outer ring shows the set amount; the inner ring shows the correction currently applied. The difference (withheld) shows where correction is deliberately held back.



Simple view: large ring and two main knobs

Ease of use

Three views

Animated transitions between Simple, Standard, and Advanced; Advanced view reveals Natural's fine-tuning controls.

Scale

75–150% sizing; separate minimum readable sizes for Retina and standard displays. The selected size is saved with the session.

Accessibility

Controls are named for screen readers; the note ring announces the target and deviation. Keyboard focus is visible.

Compatibility and performance

FEATURE	VALUE
Plug-in formats	Audio Unit (AU), VST3, AAX. Can also be built as a standalone application.
Category	Effect · Pitch Shift (AU au_fx Snvr Szio)
Operating system	macOS 11.0 and later
Processor architecture	Universal: Apple Silicon (arm64) and Intel (x86_64). Intel verified by auval, latency, tests and audio comparisons (same latency as Apple Silicon; audio difference ≤ −144 dBFS).
Channel configuration	Mono → mono, stereo → stereo
Sample rates	Natural and Synthetic: verified at 44.1 / 48 / 96 kHz. Synthetic works only at these three rates; Natural works at other rates.
Buffer size	Works at any buffer size. Output is independent of the DAW buffer size: the same input gives bit-identical output at different buffer sizes (verified by tests).
Internal processing	64-bit floating point. No memory allocation on the audio thread (verified by tests); parameter changes use a real-time-safe path.
Latency compensation	Exact latency reported for every mode and setting (see page 6). Bypass passes dry audio with the same latency; toggling bypass preserves alignment.
State and presets	One state format across all plug-in formats; compatible with older sessions and presets. Corrupt or unrecognized state data is not loaded; the current state is preserved.
DAW tests	Latency compensation verified in Logic Pro (AU), Pro Tools (AAX), and REAPER (VST3): every DAW correctly applies the reported latency at every setting.
Automated tests	About 1,900 automated tests: audio engines, latency, streaming, parameters, interface layout, pixel-level visual consistency, and accessibility.

CPU load

Apple M1 (Mac mini), single core: audio block processing time as a fraction of block duration. Average and worst 0.1%; product limits are 25% on average and 70% in the worst slice. All 46 of 46 test cases passed, with no audio dropouts.

MODE / SETTING	48 kHz AVERAGE	96 kHz AVERAGE	WORST 0.1%
Natural	≈ 4.7%	≈ 10.2%	Up to 31%
Natural, transposed	≈ 4.7–4.8%	≈ 10.2–10.4%	Up to 31%
Natural, Voice Range 80 Hz	≈ 4.1%	≈ 8.8–9.0%	Up to 23%
Natural, all advanced controls active	≈ 4.7%	≈ 10.2–10.3%	Up to 34%
Natural, Formant knob active	≈ 6.1%	≈ 13.2%	Up to 32%
Synthetic (all settings)	≈ 3.2%	≈ 6.7%	Up to 29%

Measurement: version 1.0.0, 64- and 512-sample buffers, a real vocal recording; each case run three times on an otherwise idle system (median). CPU load was not measured separately on Intel Macs.

Limitations and known issues

The following conditions have been measured and documented; they are listed to clearly show where the product works best.

Usage limitations

For monophonic vocals. Sonavyr tracks one pitch at a time; it is not designed to correct chords, layered vocals, or a mix of instruments.

Natural's latency. Approximately 79–93 ms (130–144 ms with positive Transpose). Suitable for recording, editing, and mixing; Synthetic is recommended for live monitoring.

Three parameters cannot be automated. Mode, Transpose, and Voice Range change latency, so DAW automation is unavailable; change them manually or with a preset.

Brief silence during latency changes. When Mode or Voice Range changes, or Transpose crosses 0, there is brief silence as the engine fills to its new latency (see page 6).

Formant works only at Transpose 0. When transposed, formants always shift with pitch; there is no option to change pitch while keeping formants fixed.

Synthetic's sample rates. Synthetic works at 44.1 / 48 / 96 kHz; Natural is used at other rates.

Known issues

- **Transposition timing in Natural:** transposed audio shifts slightly against reported latency: up to ≈ 6 ms early at +5, up to 3 ms late at -5; more at larger transpositions. DAW latency compensation works correctly.
- **Synthetic, 96 kHz:** audio starts 1 sample (0.01 ms) later than the reported latency.
- **Synthetic's range extremes:** in very low male rap and very high/fast sung passages, correction may fall short of the intended accuracy.
- **Synthetic, Transpose -12:** gaps may occur in parts of very high-pitched audio (e.g. high female vocals). This does not occur at Transpose -5 or above.
- **Formant knob and head voice:** formant shifting may give very high head-voice notes an unusual timbre.
- **Formant knob and stereo:** at ± 5 , very slight widening is measured in stereo recordings (left/right level difference up to 0.6 dB); it is not expected to be audible.

Release status

This document describes version 1.0.0. The release is universal: Apple Silicon and Intel Macs. The Intel binary was validated on Apple Silicon (Rosetta); it has not yet been tested on a physical Intel Mac, and Intel CPU load has not yet been measured. Apple Developer ID signing and notarization, and Avid/PACE signing for AAX, will be completed for the distribution release.

All values in this document come from version 1.0.0's code, latency table, and recorded measurements. CPU load was measured on Apple M1; it may differ on other systems.